Module Review: Bacterial Purine Base Oxidation to Urate in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG ppu00230 "Purine metabolism" (module area: nucleotide_metabolism) **Module under review:** *Bacterial purine base oxidation to urate* — guanine and hypoxanthine converge on xanthine, which is oxidized to urate. **Date:** 2026-08-19

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1. Executive summary

The purine-base oxidation-to-urate module is **present and functional** in *P. putida* KT2440. The organism uses adenine, guanine, hypoxanthine, xanthine and uric acid as **sole nitrogen sources**, direct strain-level evidence from the McpH chemoreceptor study (P 26355499).

All catalytic steps of the module are encoded within a single contiguous purine-catabolic gene island (**PP_4278–PP_4290**):

- **Terminal step (xanthine → urate)** and **hypoxanthine → xanthine**: the two-subunit NAD^+-dependent xanthine dehydrogenase **XdhAB** (PP_4278 / PP_4279, EC 1.17.1.4), matured by the accessory factor **XdhC** (PP_4280, *absent from candidate metadata*).
- **Guanine → xanthine**: guanine deaminase **guaD** (PP_4281, EC 3.5.4.3).

Two curation-critical issues: (1) **PaoABC (PP_3308–PP_3310)** is bucketed in ppu00230 but is a **periplasmic aldehyde oxidoreductase**, not a xanthine oxidase — a likely over-propagated annotation that must **not** be counted toward the terminal step. (2) **xdhC (PP_4280)** is a required accessory factor missing from the candidate list. The module boundary should stop at urate; the co-localized HIU/allantoin genes (PP_4285–PP_4288) and uric-acid permease (uacT, PP_4290) belong to the **downstream urate-degradation module**.

Overall satisfiability: **COVERED** for all three module steps, with high confidence.

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2. Target-organism pathway definition

Included biochemistry (module scope): - Guanine + H_2O → xanthine + NH_3 (guanine deaminase, Zn-dependent amidohydrolase; EC 3.5.4.3) - Hypoxanthine + NAD^+ + H_2O → xanthine + NADH (xanthine dehydrogenase; EC 1.17.1.4) - Xanthine + NAD^+ + H_2O → urate + NADH (xanthine dehydrogenase; EC 1.17.1.4)

Neighboring processes to keep separate: - De novo purine biosynthesis / IMP–AMP–GMP interconversion and salvage (purF, purL, purM, purN, purH, purD, purC, purK, purE, purB, purA, guaA, guaB, apt, xpt, hpt/PP_0747, prs, gmk, adk, ndk, nrdAB, etc.). These dominate the 65-gene candidate list but are **anabolic/salvage** functions, **not** catabolic oxidation to urate. Exclude from this module. - **Urate degradation to allantoin / glyoxylate + urea** (urate oxidase → HIU → OHCU → allantoin → allantoate → ureidoglycolate; pucM PP_4285, pucL PP_4287, puuE PP_4286, allA PP_4288, allE PP_3530, PP_4310, ureABC PP_2843-5). This is the **next** module downstream of urate. - **Adenine deamination to hypoxanthine** (adenine deaminase PP_0591, EC 3.5.4.2) feeds the hypoxanthine branch but sits one step **upstream** of the module boundary (module begins at guanine/hypoxanthine). - **Nucleotide/nucleoside catabolism upstream of free bases** (5′-nucleotidases, purine nucleoside phosphorylases ppnP/yfiH/deoD) — feeder reactions, separate.

Alternate names / database definitions: KEGG lumps all of the above into a single overview map ppu00230 "Purine metabolism"; the oxidation-to-urate segment corresponds to the classic **purine catabolism / purine dissimilation** pathway (MetaCyc "purine nucleobases degradation", "guanine and adenine ring degradation"). EC 1.17.1.4 (NAD^+-dependent xanthine **dehydrogenase**) should be distinguished from EC 1.17.3.2 (O_2-dependent xanthine **oxidase**) — KT2440 is annotated as the dehydrogenase.

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3. Expected step model and satisfiability calls

Module step	Reaction	KT2440 gene(s)	Call
Guanine supply	guanine → xanthine	**guaD** PP_4281 (EC 3.5.4.3)	**covered**
Hypoxanthine supply	hypoxanthine → xanthine	**xdhA/xdhB** PP_4278/PP_4279 (EC 1.17.1.4), + xdhC PP_4280	**covered**
Terminal oxidation	xanthine → urate	**xdhA/xdhB** PP_4278/PP_4279 (EC 1.17.1.4), + xdhC PP_4280	**covered**
(accessory)	MoCo maturation/ sulfuration for Xdh	**xdhC** PP_4280	covered (add to module)

Not part of this module (keep separate / boundary): urate → HIU (urate oxidase; no co-clustered classical uricase identified — see §5), HIU/OHCU/allantoin genes, uacT permease, adenine deaminase.

4. Candidate genes and evidence

High-confidence, in-module

- **xdhA — PP_4278 (Q88F21), EC 1.17.1.4.** Small subunit of the two-component bacterial xanthine dehydrogenase (2Fe-2S/FAD module). Evidence: sequence/annotation + genomic context (head of the catabolic island). A single XdhAB (EC 1.17.1.4) canonically performs **both** hypoxanthine→xanthine and xanthine→urate, satisfying two module steps. Caveat: no direct KT2440 enzymology published; assignment rests on strong orthology + operon context.
- **xdhB — PP_4279 (Q88F20), EC 1.17.1.4.** Large molybdopterin (MoCo)-binding catalytic subunit (799 aa). Same evidence and caveats as xdhA. Together XdhAB is the "two-subunit XdhAB complex" of the generic module realization.
- **xdhC — PP_4280 (Q88F19), EC 1.17.1.4 (accessory).** XdhC-family maturation factor (MoCo sulfuration / cofactor insertion), immediately downstream of xdhB. **Not in the candidate metadata** — should be added to the module as a required accessory (non-catalytic) component. Promote to full review.

- **guaD — PP_4281 (Q88F18), EC 3.5.4.3.** Guanine deaminase (guanine aminohydrolase), the guanine-supply branch to xanthine. Adjacent to xdhABC in the island. UniProt confirms it **binds 1 Zn^{2+} per subunit** (metallo-dependent hydrolase superfamily), **exactly matching the generic module's "zinc-dependent guanine deamination"** realization. Evidence: orthology + operon context + cofactor annotation; no direct KT2440 assay found but assignment strong.

Direct physiological support (strain-level)

- **McpH — PP_2643 (not in candidate list; chemoreceptor).** Binds adenine, guanine, xanthine, hypoxanthine, uric acid; enables their use as sole N sources (P 26355499). Confirms flux through the module in vivo. Not an enzyme of the module but the strongest **direct KT2440 evidence** that the pathway operates.

In bucket but OUTSIDE this module (boundary/feeder)

- **PP_0591 adenine deaminase (EC 3.5.4.2)** — adenine → hypoxanthine; upstream feeder.
- **pucM PP_4285, pucL PP_4287, puuE PP_4286, allA PP_4288, allE PP_3530, PP_4310** — downstream urate/allantoin degradation.
- **uacT PP_4290 (uric acid permease)** — urate uptake; module-boundary transporter (missing from candidate list).
- **De novo/salvage purine genes** (purF/L/M/N/H/D/C/K/E, purA/B, guaA/B, apt, xpt, PP_0747 hpt, prs, gmk, adk, ndk, nrdAB, ppnP, amn, PP_3662, dgt, PP_5100, cyaA, relA/spoT, ushA, surE, yrfG, nudE/F, apaH, mazG, ppx, etc.) — anabolic, salvage, signalling, or generic nucleotide housekeeping; **not** oxidation-to-urate.

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5. Gaps, ambiguities, and likely over-annotations

1. **PaoABC (PP_3308/09/10) — likely over-propagated into ppu00230.** Annotated "promiscuous aromatic aldehyde dehydrogenase (EC 1.2.99.7)." The *E. coli* ortholog is the **periplasmic aldehyde oxidoreductase** — a molybdopterin-cytosine-dinucleotide heterotrimer of the xanthine-oxidase **structural** family whose physiological role is **aldehyde detoxification**, with a surface-exposed active site lacking aromatic residues

(P 27622978 ; P 24492481 ; Tat-exported heterotrimer, P 25531212).

Reinforced by **genomic context**: in KT2440 the paoABC operon (subunit sizes PaoA 175 aa / PaoB 333 aa / PaoC 738 aa, matching *E. coli*) is flanked by Rho termination factor (PP_3307), a K^+/H^+ antiporter (PP_3311) and a heat-shock protein (PP_3312) — **no purine-catabolic genes nearby**. **Do not count toward xanthine→urate.** Recommend mark as **not-in-module / mis-bucketed** and flag EC 1.2.99.7 (not 1.17.1.4).

2. **xdhC (PP_4280) missing from candidate metadata** — required accessory factor; add to module.
3. **uacT (PP_4290) missing** — uric acid permease at the module boundary.
4. **Downstream uricase identified (gap closed).** No classical uricase is *co-clustered* with xdhABC, but a proteome search identifies **puuD (PP_3099), "Uricase/urate oxidase" (EC 1.7.3.3), 500 aa**, which performs urate → 5-hydroxyisourate — the **first step of the NEXT (urate-degradation) module**, feeding pucM/pucL. No enterobacterial-type HpxO FAD-monooxygenase or HpxDE Rieske system is present, consistent with the conventional molybdoflavo route. This confirms the terminal boundary of the oxidation-to-urate module sits cleanly at urate, with PuuD initiating downstream catabolism.
5. **Lineage-specific alternative realization exists.** In enterobacteria (e.g., *Klebsiella oxytoca/ pneumoniae*) (hypo)xanthine→urate is done by a **Rieske two-component oxygenase HpxDE**, not by molybdoflavo XdhAB (P 19060149). KT2440 uses the **conventional molybdoflavo XdhAB** route; the generic "two-subunit XdhAB" realization fits. Transfer of any Hpx-based annotation to KT2440 would be **incorrect**.
6. **No direct KT2440 enzymology** for XdhAB or GuaD substrate range; assignments are homology + genomic-context + physiology (growth on purines). Confidence high but not biochemically proven in-strain.

6. Module and GO-curation recommendations

- **Mark COVERED:** guanine→xanthine (guaD PP_4281); hypoxanthine→xanthine (XdhAB PP_4278/79); xanthine→urate (XdhAB PP_4278/79). Confidence: high (orthology + operon + physiology).
- **Add to module: xdhC PP_4280** as accessory/maturation factor (GO:0043545 molybdopterin cofactor metabolic/insertion context; associate with XdhAB assembly). Consider adding **uacT PP_4290** as the associated urate transporter (module boundary).
- **Exclude / re-bucket: PaoABC PP_3308-3310** — remove from the purine-oxidation module; annotate as periplasmic aldehyde oxidoreductase (EC 1.2.99.7). Recommend flag as **over-propagated** in ppu00230.
- **Boundary correction:** the generic module correctly ends at urate. Keep the HIU/OHCU/allantoin genes and ureases in the separate **urate/allantoin degradation** module; do not let ppu00230's overview scope merge them.
- **GO term adequacy:** existing GO terms suffice — GO:0004855 (xanthine dehydrogenase activity)/GO:0004854 (xanthine oxidase) for XdhAB, GO:0008892 (guanine deaminase activity) for guaD. No new GO requests appear necessary. A **new module document is not needed**; only metadata edits (add xdhC, drop PaoABC).
- **module_needs_revision:** minor — add xdhC, remove/flag PaoABC.

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7. Genes to promote to full `fetch-gene` review

1. **xdhC — PP_4280** (missing accessory factor; confirm role and module membership).
2. **PaoABC — PP_3308/PP_3309/PP_3310** (resolve mis-bucketing; confirm EC 1.2.99.7 aldehyde-oxidoreductase function, exclude from module).
3. **guaD — PP_4281** and **xdhA/xdhB — PP_4278/PP_4279** (confirm substrate range; ideally direct evidence that a single XdhAB does both hypoxanthine and xanthine oxidation in KT2440).
4. **Cryptic urate oxidase** search (downstream): identify the uricase feeding pucM/pucL, since none is co-clustered.

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8. Key references

- Fernández M, Morel B, Corral-Lugo A, Krell T. *Identification of a chemoreceptor that specifically mediates chemotaxis toward metabolizable purine derivatives.* Mol Microbiol. 2016.

  P 26355499

  — Direct KT2440 evidence: adenine, guanine, xanthine, hypoxanthine, uric acid used as sole N sources.
- Correia MAS, et al. *The Escherichia coli Periplasmic Aldehyde Oxidoreductase Is an Exceptional Member of the Xanthine Oxidase Family of Molybdoenzymes.* J Biol Chem/Biochemistry 2016.

  P 27622978

  — PaoABC is an aldehyde oxidoreductase, not a xanthine oxidase.
- Otrelo-Cardoso AR, et al. *Structural data on the periplasmic aldehyde oxidoreductase PaoABC…* 2014.

  P 24492481

  — PaoABC role = aldehyde detoxification.
- Lee PA, et al. *…natural three-component hitchhiker mechanism (PaoABC Tat export).* 2014.

  P 25531212

  — PaoABC is a periplasmic (Tat-exported) heterotrimer.
- Pope SD, Chen L-L, Stewart V. *Purine utilization by Klebsiella oxytoca M5al: genes for ring-oxidizing and -opening enzymes.* J Bacteriol. 2009.

  P 19060149

  — Enterobacterial Rieske HpxDE alternative vs. conventional molybdoflavo xanthine dehydrogenase.
- UniProt (proteome UP000000556) locus annotations PP_4277–PP_4291 — purine-catabolic gene island (xdhABC, guaD, pucM/puuE/pucL/allA, uacT).

Confidence & species-transfer notes

- **Direct KT2440 experimental:** growth on purines / N-source use and chemotaxis (P 26355499).
- **Homology + genomic context (strong transfer):** XdhAB, XdhC, GuaD identities and operon structure (UniProt/KEGG for PSEPK itself — not cross-species).
- **Comparative (function-defining, from related taxa):** PaoABC function (*E. coli*); Rieske HpxDE alternative (*Klebsiella*) — used to *exclude* mis-annotations, transfer of the negative/alternative is appropriate.
- **Not established in-strain:** purified-enzyme kinetics of KT2440 XdhAB/GuaD; identity of the downstream uricase.

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